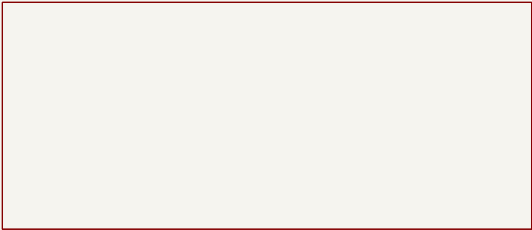


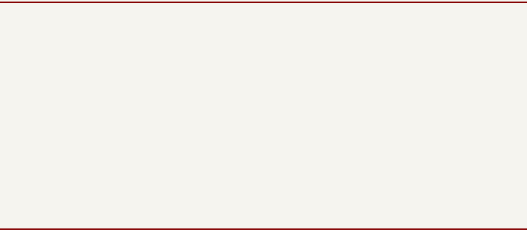
Vanchor Helm Board – sheet index

- 1 carrier: 12V in, protection, 5V buck header, Orange Pi Zero 3 ribbon header, UART/I2C JSTs
- 2 control: Pico 2, external thrust–driver interface (J13), BTN8982 servo bridge, AS5600

carrier



control



12V battery only; 5V from generic buck module on J14
Orange Pi Zero 3 (IDC ribbon) + Pico 2; external IBT–2 thrust driver; on–board servo bridge

vanchor–ng

Sheet: /
File: vanchor–helm.kicad_sch

Title: Vanchor Helm Board

Size: A3Date: 2026–07–02

Rev: v4.1

KiCad E.D.A. 10.0.3

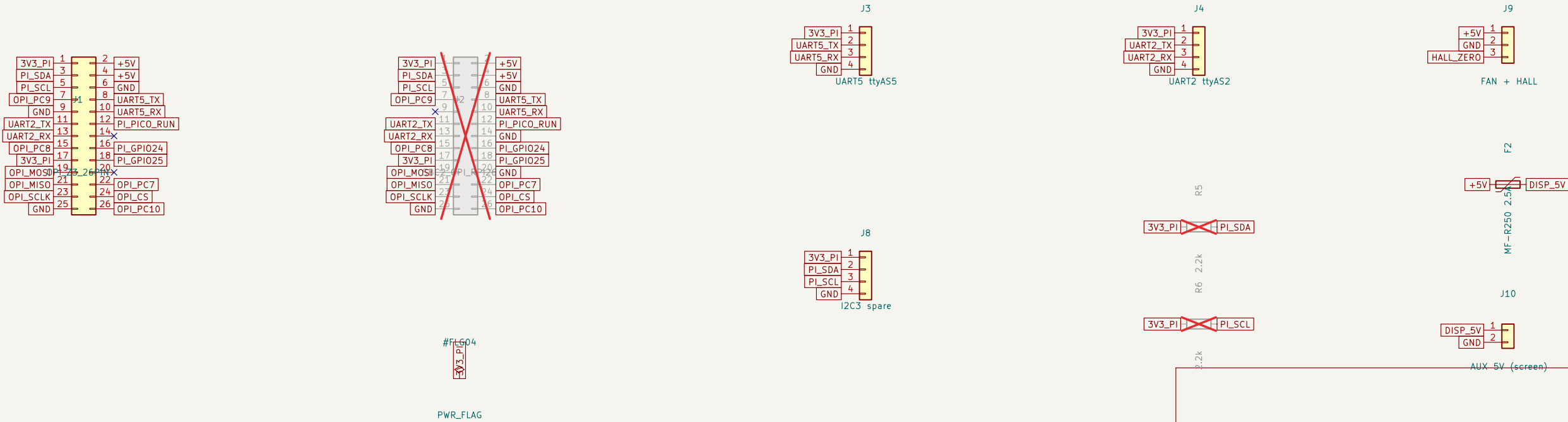
Id: 1/3

POWER: 12V battery -> F1 -> reverse-FET -> VIN; XL4015 buck daughterboard U5 (set 5.1V first)

CARRIER: J1 -> 26-way IDC ribbon -> Orange Pi Zero 3 (powered through the ribbon, no USB-C)



ORANGE PI ZERO 3 HEADER + UART/I2C BREAKOUTS



PICO 2: I2C slave 0x42 to the Zero 3; SWD reflash via ribbon pins 16/18, RUN on 12

THRUST: J13 = IBT-2 pin order to the external driver; EN pull-downs = failsafe off

SERVO: 2x BTN8982TA on protected VIN; AS5600 encoder cable on J11

THRUST DRIVER INTERFACE

